

UDF

Summary

Amazon Redshift UDFs allow creation of custom reusable functions using SQL or Python. SQL UDFs are ideal for simple query-based logic, while Python UDFs support complex processing. UDFs can be marked as VOLATILE, STABLE, or IMMUTABLE depending on result behavior, and using IMMUTABLE functions improves performance through caching.

Creating UDF

```
CREATE [ OR REPLACE ] FUNCTION f_function_name
( { [py_arg_name py_arg_data_type |
sql_arg_data_type ] [ , ... ] } )
RETURNS data_type
{ VOLATILE | STABLE | IMMUTABLE }
AS $$
{ python_program | SELECT_clause }
$$ LANGUAGE { plpythonu | sql }
```

UDF: SQL

Return the greater value between two numbers.

```
create function f_sql_greater (float, float)
returns float
stable
as $$
select case when $1 > $2 then $1
else $2
end
$$ language sql;
```

SQL UDF Parameters

Parameters are referenced using:

```
$1, $2, $3
```

Based on parameter order.

UDF: Python

Same logic using Python syntax.

```
create function f_py_greater (a float, b float)
returns float
stable
as $$
if a > b:
    return a
```

```
return b
$$ language plpythonu;
```

Useful for:

- Mathematical operations
- String manipulation
- Loops
- Complex conditions
- Advanced transformations

Advantage

Variables can use actual names:

```
a, b
```

Unlike SQL positional parameters.

Volatility

Volatility tells Redshift how often function results may change.

```
CREATE [ OR REPLACE ] FUNCTION f_function_name
( { [py_arg_name py_arg_data_type |
sql_arg_data_type ] [, ... ] } )
RETURNS data_type
{ VOLATILE | STABLE | IMMUTABLE }
AS $$
{ python_program | SELECT_clause }
$$ LANGUAGE { plpythonu | sql }
```

1. VOLATILE

Behavior

Can return different results every time.

Example

Random number generator.

```
VOLATILE
```

2. STABLE

Behavior

Result stays same within one query execution.

Example

Current date function.

STABLE

3. IMMUTABLE

Behavior

Always returns same output for same input.

Example

Addition or total calculation.

IMMUTABLE

Benefit

Improves query performance because Redshift can cache results.